Efficiency decomposition, aggregation sensitivity and reproducibility in a finite-lattice overrelaxation study

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Abstract

We analyse the paired overrelaxation-efficiency campaign of the Yang–Mills Mass Gap Laboratory of Elsbernd. The per-pair ESS-per-second ratio factors exactly into a decorrelation gain and a cost factor; in all 27 paired conditions the cost penalty (2.8–8.4×) outweighed the gain (at most 3.11×), so no ratio exceeded one (maximum 0.94). We show that two natural aggregation rules disagree, that τ_{int} is deterministic while ESS/s is not, and report a cross-platform re-run in which hybrid chains did not reproduce. This is a finite-lattice methodology study and makes no statement about the continuum mass gap.

1 Setting

The study uses the lattice and campaign of Elsbernd, *Yang–Mills Mass Gap Laboratory* `https://github.com/Elsbernd09/yang-mills-mass-gap-lab`: SU(2) Wilson action on a $5 \times 4 \times 4$ lattice, $\beta \in \{1.5, 2, 2.5\}$, seeds 2026, 2027, 2028, 180 measured updates per chain, and the schedules M, M+1OR, M+2OR, M+4OR (one Metropolis sweep followed by 0, 1, 2, 4 microcanonical overrelaxation sweeps). The campaign and its manifests are Elsbernd's; the analysis below only post-processes the recorded results.

2 Symbolic models

The following relations were derived or verified with `sympy` (`ymlab.symbolic`) and are numerically cross-checked against the laboratory code in the test-suite.

Lattice coupling. Wilson coupling for SU(N).

$$\beta = \frac{2N}{g^2} \tag{1}$$

Bare coupling from beta. bare coupling solved from the Wilson coupling relation.

$$g = \frac{\sqrt{2}\sqrt{N}}{\sqrt{\beta}} \tag{2}$$

SU(2) plaquette term. for a unit quaternion, $\det U = a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1$.

$$w_p = 1 - a_0 \tag{3}$$

Effective sample size. ESS from integrated autocorrelation time.

$$N_{eff} = \frac{n_{meas}}{2\tau} \tag{4}$$

Efficiency ratio. paired ESS-per-second ratio; independent of the number of measurements.

$$R = \frac{t_M \tau_M}{t_{OR} \tau_{OR}} \tag{5}$$

Break-even. Metropolis autocorrelation time at which the efficiency ratio equals one.

$$\tau_M = \frac{t_{OR} \tau_{OR}}{t_M} \tag{6}$$

AR(1) integrated autocorrelation time. geometric sum of autocorrelations for $|\varphi| < 1$.

$$\tau_{int} = \frac{\varphi + 1}{2 - 2\varphi} \tag{7}$$

Cosh effective mass. the estimator ratio $[C(t-1)+C(t+1)]/2C(t)$ equals $\cosh(m)$, independent of t.

$$m_{eff} = \operatorname{acosh}(\cosh(m)) \tag{8}$$

Creutz area law. perimeter and constant terms cancel in the Creutz ratio.

$$\chi = \sigma \tag{9}$$

O(N) mass log-slope. logarithmic slope of the O(N) mass against β at one loop plus the prefactor power.

$$s_{O(N)} = -\frac{2\pi}{N-2} + \frac{1}{b\,(N-2)} \tag{10}$$

SU(N) spacing log-slope. logarithmic slope of the lattice spacing against $\beta = 2N/g^2$ from the two-loop beta function.

$$s_{SU(N)} = \frac{51}{121 b} - \frac{12\pi^2}{11 N_c^2} \tag{11}$$

One-loop coefficient (non-SUSY). SU(N) with n_f Dirac fundamentals; $b < 0$ is asymptotic freedom.

$$b = -\frac{11N}{3} + \frac{2n_f}{3} \tag{12}$$

One-loop coefficient (SUSY). SU(N) with n_f chiral flavour pairs; $b < 0$ is asymptotic freedom.

$$b = -3N + n_f \tag{13}$$

3 Efficiency decomposition

Across all 27 pairs the plaquette ratio lay in [0.05, 0.94]; the gain exceeded one in 12 pairs (maximum 3.11) while the cost penalty lay in [2.8, 8.4]. For this implementation the additional sweeps were not repaid.

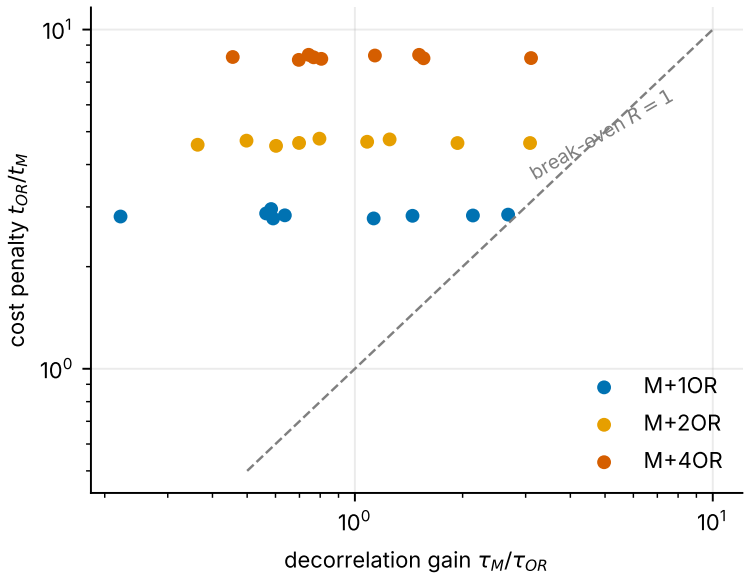


Figure 1: Per-pair decorrelation gain against cost penalty (log axes). Points below the dashed line would have $R > 1$; none do.

4 Aggregation sensitivity

The mean of paired ratios and the ratio of mean ESS/s are different statistics. They differ by up

β	schedule	mean gain	mean cost factor	mean of ratios	ratio of means	pair range
1.5	M+1OR	0.93	0.357	0.333	0.281	0.08–0.51
1.5	M+2OR	1.08	0.217	0.234	0.207	0.13–0.42
1.5	M+4OR	1.02	0.122	0.124	0.114	0.09–0.19
2	M+1OR	1.10	0.354	0.388	0.263	0.20–0.75
2	M+2OR	1.32	0.216	0.284	0.152	0.08–0.67
2	M+4OR	1.44	0.121	0.175	0.103	0.05–0.38
2.5	M+1OR	1.30	0.347	0.454	0.404	0.20–0.94
2.5	M+2OR	1.04	0.212	0.221	0.227	0.17–0.26
2.5	M+4OR	1.13	0.119	0.134	0.133	0.09–0.18

Table 1: Aggregates per (β, schedule), three seeds each.

to 86% of the ratio of means.

5 Determinism and reproduction

Two published campaigns with identical configuration hash agree in τ_{int} to 0.0e+00 (maximum) and in ESS/s to within 3.9%.

On Linux x86_64 (Python 3.13.18, NumPy 2.5.3) the published seed-2026 conditions were re-run; the published runs were made on Darwin arm64 (Python 3.9.6, NumPy 2.0.2). The pure-Metropolis chains reproduce the published τ_{int} to within 2.2e-13; 0 of 9 hybrid chains reproduce it exactly, with local-to-published τ_{int} ratios between 0.88 and 4.12. Median runtime here was

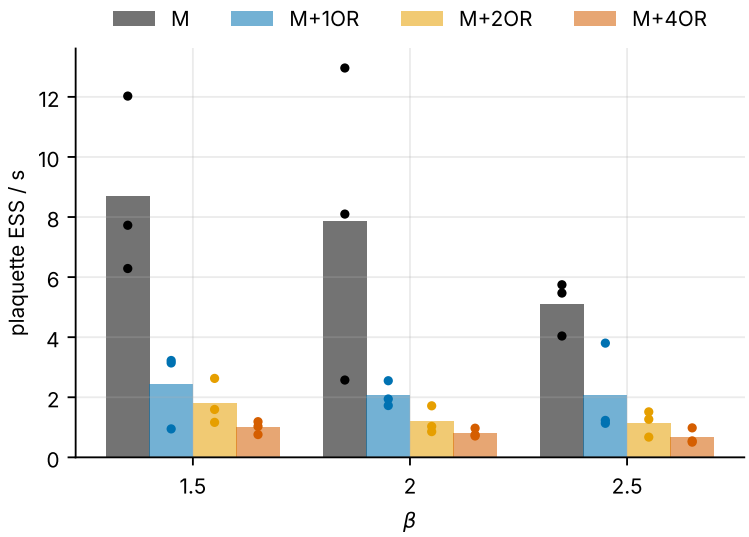


Figure 2: Plaquette ESS/s by schedule and *beta*; dots are individual seeds.

2.2× the published runtime. The cause of the hybrid divergence was not isolated (platform, library version, and floating-point ordering are all confounded).

6 Asymptotic freedom: two-dimensional O(N) against four-dimensional SU(2)

The physics tools of the OpenAI-math fork (`https://github.com/brentharts/openai_math_yang_mills`, commit `ba56ef3eb`) include a Wolff cluster Monte Carlo of the two-dimensional O(N) model, whose output is stored with the laboratory. The leading logarithmic slope of the mass (O(N)) or lattice spacing (SU(N)) against β is derived with sympy; for SU(N) it is $-12\pi^2/(11N^2)$ plus a correction $51/(121\beta)$ independent of N.

O(N). O(3): measured/predicted local slope 0.64, 0.75, 0.86 over xi = 3.4 to 11.8 (smallest L/xi 8.8); O(4): measured/predicted local slope 0.89, 0.89, 0.95 over xi = 3.0 to 14.6 (smallest L/xi 7.9). The measured slope approaches the asymptotic form from below as β grows, which needed correlation lengths of order ten lattice spacings. Halving the scale takes $\Delta\beta \approx 0.12$ for O(3), 0.24 for O(4), 0.27 for SU(2) and 0.62 for SU(3) at the couplings named in the text file.

SU(2). 4^4 lattice: measured/two-loop slope 0.74, 0.35, 0.17, 0.22, 0.20 at $\beta = 2.1, 2.3, 2.5, 2.7, 2.9$; 6^4 lattice: measured/two-loop slope 0.37, 0.55, 0.22, 0.24, 0.14 at $\beta = 2.1, 2.3, 2.5, 2.7, 2.9$. The SU(2) lattices are far from asymptotic scaling: the slope of ln $\sqrt{\chi}$ is 74% of the two-loop value at best, and at most 24% over the last three βs of every lattice (at the largest β the ratio is 0.20 on 4^4 and 0.14 on 6^4, so the larger lattice did not move it toward one), a possible reason is strong-coupling and finite-size effects in $\chi(2,2)$ on a lattice this small, which was not tested here. The plaquette, by contrast, is already close to its weak-coupling leading term at β of 2.8 and above (4^4: largest —plaquette - (1 - 3/4β)— at β ¿= 2.8 is 0.032; 6^4: largest —plaquette - (1 - 3/4β)— at β ¿= 2.8 is 0.031). The O(N) models reach 0.86 to 0.95 of the asymptotic slope at their largest

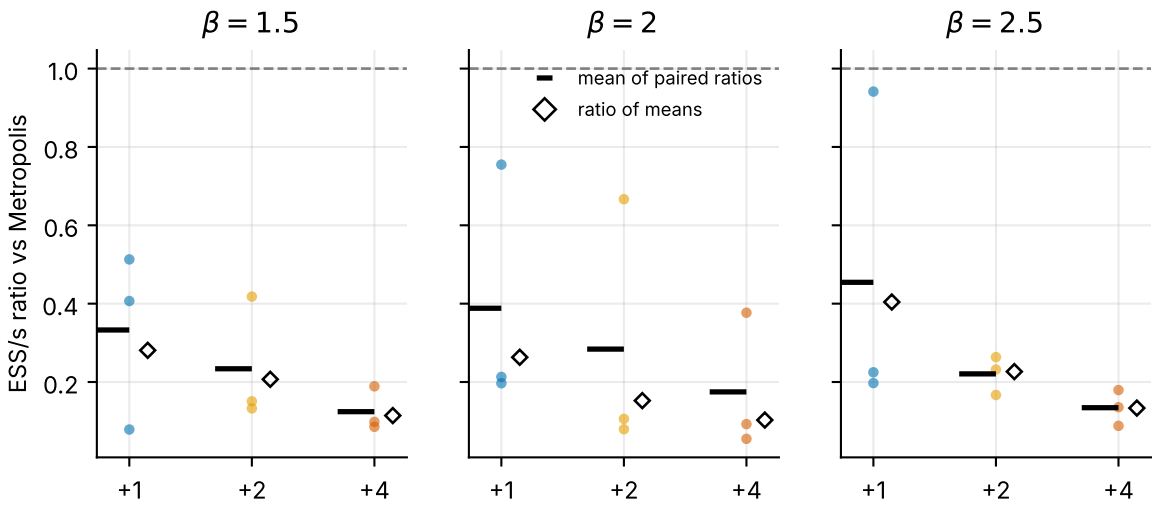


Figure 3: Per-seed ratios with the two aggregates; dashed line is break-even.

β (correlation lengths up to 15 lattice spacings); the SU(2) scans are not run at lattice sizes where a correlation length could be measured at all.

These comparisons are descriptive: the O(N) models are a toy for Yang–Mills, the SU(2) lattices are tiny with one chain per β, and nothing here concerns the continuum mass gap.

7 One-loop cross-check with CICY

The CICY package (`https://github.com/brentharts/CICY`, `pyCICY.theories.running`) computes one-loop coefficients b_i from an index and uses b_3 in the QCD scale $\Lambda = M \exp(-2\pi/(|b_3|\alpha))$. With $d(1/\alpha)/d\ln\mu = b/2\pi$, sympy gives $b = -\frac{11}{3}N + \frac{2}{3}n_f$ (Dirac fundamentals) and $b = -3N + n_f$ (supersymmetric pairs). For $N = 3$, $n_f = 2n_g$ this reproduces SU(3) b_3: $n_g = 3$: -3, $n_g = 4$: -1, $n_g = 5$: $+1$, the values CICY states, and $b_3 = -7$ for the Standard Model; asymptotic freedom needs $n_f < 3N$, at most four generations. The exponent $2\pi/(|b|\alpha)$ equals the lattice-spacing exponent $1/(2b_0 g^2)$ with $b_0 = -b/16\pi^2$, which is $11N/48\pi^2$ at $n_f = 0$. This compares formulas with formulas; it adds no evidence about the lattice data.

8 Limits

Three seeds, one lattice, SU(2), plaquette emphasised; the ranking depends on the Python/NumPy cost structure. The cross-platform divergence is observed, not explained.

Acknowledgement. All simulation code, the campaign design and the recorded data are the work of Michael Elsbernd; this document is an analysis layer on top of that work.

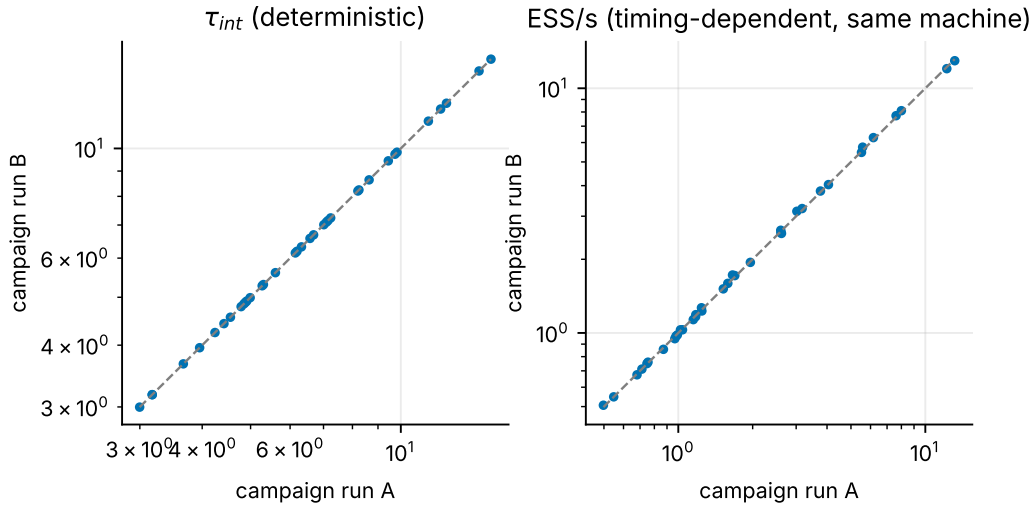


Figure 4: Run-to-run agreement on one machine.

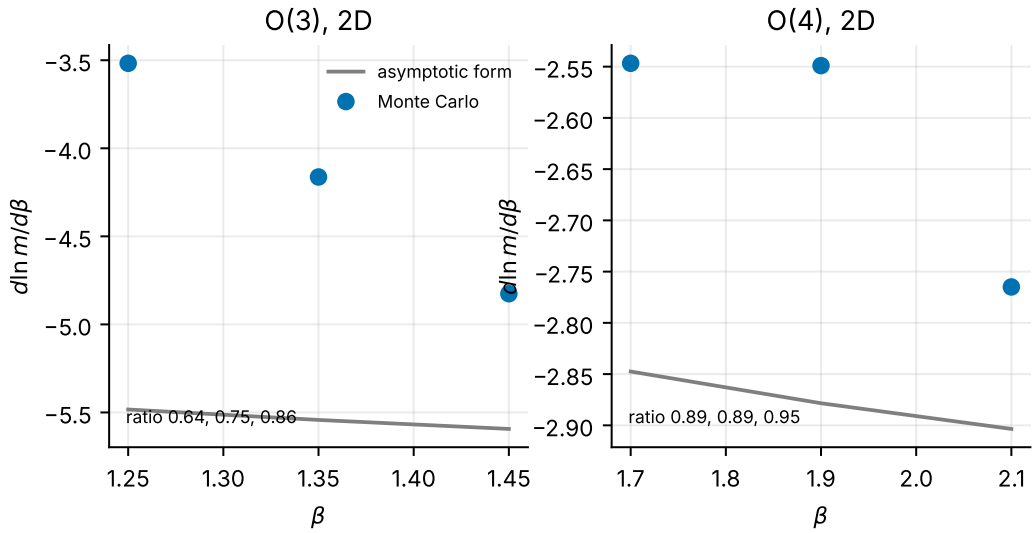


Figure 5: Measured local slope of $\ln m$ against β and the asymptotic form.

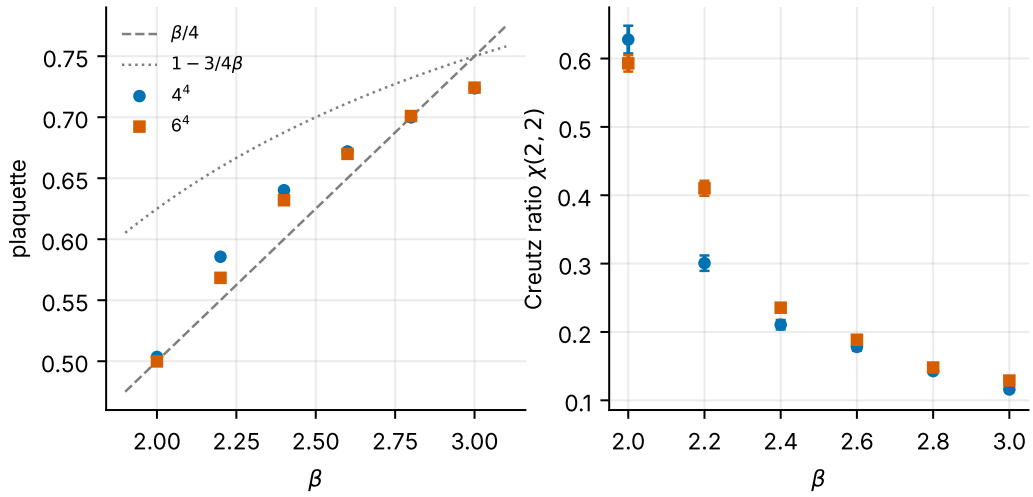


Figure 6: Plaquette and Creutz ratio from the SU(2) scans; dashed and dotted lines are the leading strong- and weak-coupling terms.

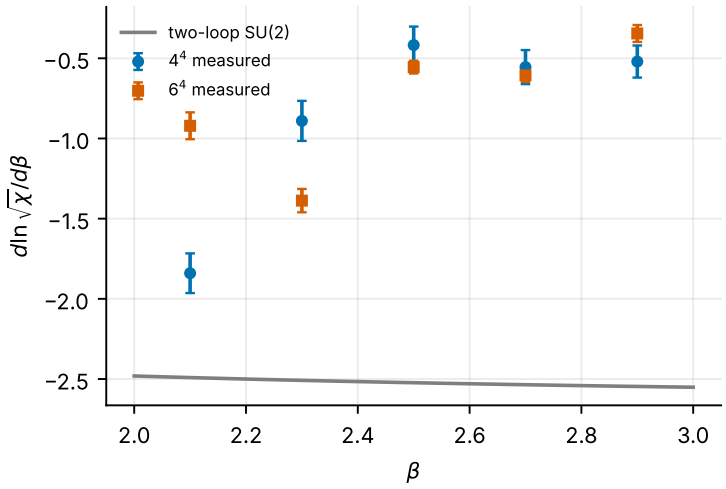


Figure 7: Measured slope of $\ln\sqrt{\chi}$ against the two-loop SU(2) slope.